



Git

Hands-On 3





Description:

This hands-on lab covers branching and merging workflows, including creating a new branch, making changes, listing local and remote branches, visualizing differences using P4Merge, merging back into the master branch, and cleaning up by deleting the branch.

WorkFlow

Step 1: Branching

Create a New Branch: Create a new branch named GitNewBranch:

git branch GitNewBranch

List Branches: List all local and remote branches to view the available branches and observe the * mark denoting your currently active branch:

git branch -a

Switch to the New Branch: Switch your working context to GitNewBranch:

git checkout GitNewBranch

Add Files and Content: Create a new file with content inside the branch:

echo "Branching and merging content" >> branch_file.txt

Stage and Commit Changes: Stage the new file and commit your changes to the branch:

git add branch_file.txt

git commit -m "Add branch_file.txt to GitNewBranch"

Check Status: Verify the status of your working directory:

git status



WorkFlow

Step 2: Merging

Switch to Master: Switch back to the master branch:

git checkout master

List Differences (CLI): Inspect differences between the master trunk and the branch via the command line interface:

git diff master GitNewBranch

List Differences (P4Merge): View visual differences between master and the branch using the P4Merge tool:

git mergetool

Merge Branch to Trunk: Merge GitNewBranch into the master branch:

git merge GitNewBranch

Observe Merge History: View a graphical and concise history log to check the merge commit:

git log --oneline --graph --decorate

Delete Branch and Verify Status: Delete the feature branch after merging and check the repository status:

git branch -d GitNewBranch

git status



Git Bash & Output

```
MINGW64:/c/Users/AADI/GitDemo

AADI@Aadi MINGW64 ~
$ cd GitDemo

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git branch -a
  GitNewBranch
* master
  remotes/origin/master

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ echo "Adding new features in the new branch" >> feature.txt
```

```
AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    feature.txt

nothing added to commit but untracked files present (use "git add" to track)

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git add feature.txt
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next time Git touches it

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Add feature.txt in GitNewBranch"
[GitNewBranch 863caa6] Add feature.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 feature.txt

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/feature.txt b/feature.txt
new file mode 100644
index 0000000..ffc7b00
--- /dev/null
+++ b/feature.txt
@@ -0,0 +1 @@
+Adding new features in the new branch
```



Git Bash & Output

```
--- /dev/null
+++ b/feature.txt
@@ -0,0 +1 @@
+Adding new features in the new branch

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 7392b8c..863caa6
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
```

```
MINGW64:/c/Users/AADI/GitDemo
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    feature.txt

nothing added to commit but untracked files present (use "git add" to track)

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git add feature.txt
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next time Git touches it

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Add feature.txt in GitNewBranch"
[GitNewBranch 863caa6] Add feature.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

AADI@Aadi MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/feature.txt b/feature.txt
new file mode 100644
index 0000000..ffc7b00
--- /dev/null
+++ b/feature.txt
@@ -0,0 +1 @@
+Adding new features in the new branch

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 7392b8c..863caa6
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 863caa6 (HEAD -> master, GitNewBranch) Add feature.txt in GitNewBranch
* 7392b8c (origin/master) Add gitignore file to ignore log files and folders
* 088c41c Welcome

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 863caa6).

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

AADI@Aadi MINGW64 ~/GitDemo (master)
```